Real-Time Quantum-Inspired 5D K-Means Image Segmentation

Sesiunea de Comunicări Științifice ale Studenților

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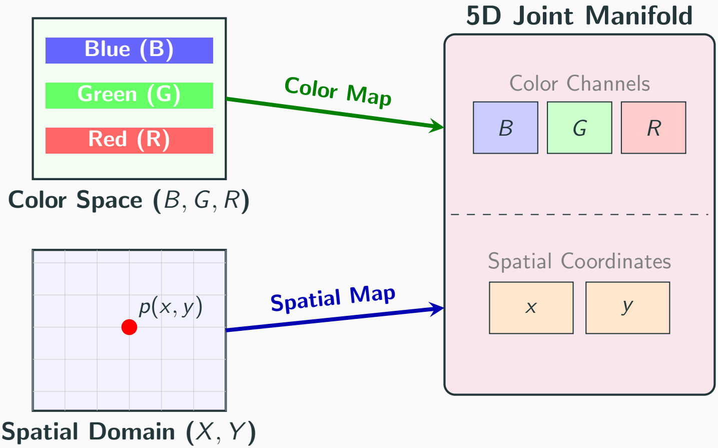
Problem Statement: The Challenge

- **Core Goal:** Achieving high-fidelity joint color-spatial (5D) image segmentation under strict real-time constraints (> 120 FPS) for live video.
- **The Segmentation Problem:** Traditional 3D color-only segmentation (e.g., in HSV or BGR) lacks spatial context, leading to highly fragmented regions.
- **5D Joint Manifold (B, G, R, X, Y):** Grouping pixels by both color and coordinates preserves spatial consistency, requiring dimension normalization and hardware-level acceleration to handle the computational overhead.

Problem Statement: 5D Joint Manifold Mapping



Problem Statement: Requirements & Bottlenecks

- **Strict Real-Time Requirement (120+ FPS):** Processing budget of < 8.33 ms per frame for interactive computer vision (e.g., robotic loops, driving simulators).
- **Hardware Bottlenecks:**
  - *CPU Limits:* Sequential 5D clustering requires millions of calculations per frame (too slow for live feeds).
  - *NISQ QPU Limits:* Real Quantum Processing Units suffer from physical noise, state preparation overhead, and massive network latency.
- **Solution:** A parallel C++/CUDA pipeline that simulates quantum state overlaps directly on local classical GPU cores.

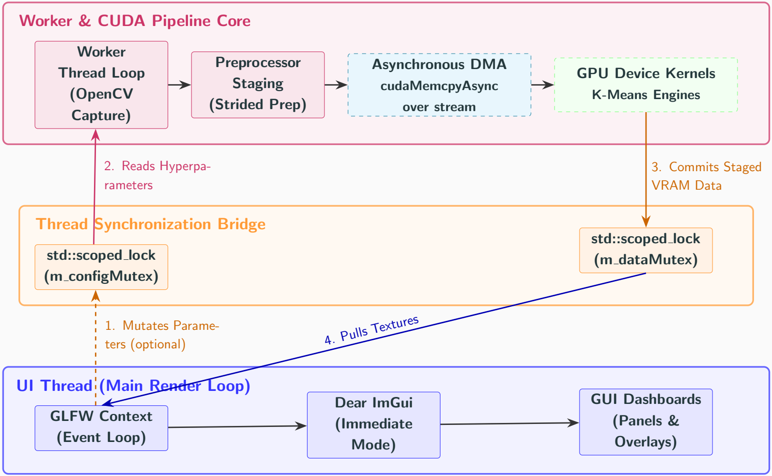
Original Contributions & What I Bring

- **Custom High-Throughput Software Package:** Fully asynchronous CUDA pipeline decoupling the UI from the computational backend.
- **Hardware-Accelerated Quantum Cosine Engine:** Custom kernel utilizing SFU math intrinsics to emulate the SWAP test in real-time.
- **Asynchronous Still-Frame Diagnostics:** On-demand diagnostics on a background thread computing the 3 key quality metrics (WCSS, Davies-Bouldin Index, and Monte Carlo-approximated Silhouette Coefficient) without GUI lag.
- **Two-Tier Atomic Reduction Kernel:** Maximized memory throughput by aggregating sub-totals in block shared memory.

System Architecture & Data Pipeline

- **Decoupled Dual-Threaded Model:**
  - *UI Thread (Main):* GLFW & Dear ImGui render loop, rendering segmented OpenGL textures asynchronously.
  - *Worker Thread:* OpenCV camera capture, preprocessor staging, and active CUDA engine execution.
- **Thread Synchronization Bridge:** Mutex-protected state machine (`std::scoped_lock`) preventing race conditions without locking the live rendering pipeline.
- **Memory Efficiency:** Page-locked (pinned) host memory and asynchronous DMA streams (`cudaMemcpyAsync`) to hide PCIe bus latency.

System Architecture Diagram



Quantum-Inspired Phase-Space Metric

- **Born's Rule & The SWAP Test:** Measuring state overlap probability:

$$P(|0\rangle) = \frac{1}{2} + \frac{1}{2}|\langle\psi|\phi\rangle|^2$$

- **Hilbert Phase Space Mapping:**
  - Mapping normalized 5D coordinate differences to angular phases in $[0, \frac{\pi}{4}]$:

$$\theta_d = |p_d - c_d| \cdot \frac{\pi}{4}$$

  - Evaluating multi-qubit product state overlaps using squared cosine curves:

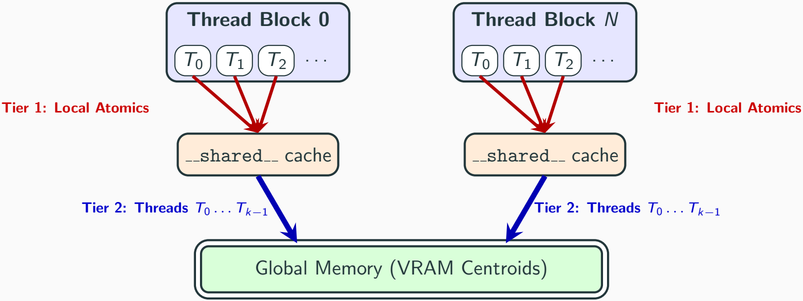
$$D_{\text{Quantum}}(\mathbf{p}, \mathbf{c}) = 1.0 - \prod_{d=1}^{D} \cos^2\left((p_d - c_d) \cdot \frac{\pi}{4}\right)$$

- **Hardware-Level Emulation:** Compiling trigonometric evaluations into fast, single-cycle device intrinsics (`__cosf`) executing on the GPU's Special Function Units (SFUs).

GPU Kernel & Memory Optimizations

- **GPU-Native Preprocessor:** Strided preprocessor formatting BGR-XY coordinates into an Array of Structures (AoS), maximizing L_1 cache line hit rates and supporting configurable stride-based spatial downsampling.
- **Templated Loop Unrolling:** Forcing compiler branch elimination via 5D templated math utilities (`#pragma unroll`) and fast math compiler flags.
- **Two-Tier Atomic Reduction:** A hierarchical aggregation strategy (detailed in the next slide) designed to minimize global memory traffic bottlenecks.

Two-Tier Atomic Reduction



- **Tier 1 (Block Atomics):** GPU threads accumulate coordinate sub-totals locally inside fast on-chip shared memory, avoiding costly global memory bus traffic.
- **Tier 2 (Global Atomics):** The first K threads per block ($T_0 \dots T_{k-1}$) flush the accumulated sub-totals to Global VRAM, parallelizing the final reduction.

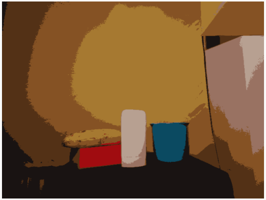
Experimental Results & Telemetry

Configuration		Classical Engine		Quantum Engine	
K (Clusters)	S (Stride)	Latency	Iterations	Latency	Iterations
3	8	1.81 ms	11.71	1.61 ms	12.38
8	4	6.45 ms	41.81	6.15 ms	41.43
40	1	26.66 ms	88.67	27.95 ms	88.62

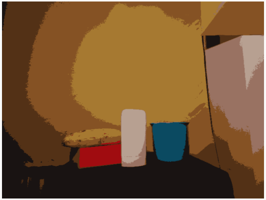
- **Speed Parity:** Dedicated GPU Special Function Units (SFUs) execute simulated trigonometric distances in a single clock cycle, maintaining latency parity with the Euclidean solver.
- **Functional Equivalence:** Although not listed in the table, all diagnostic quality metrics (WCSS, Davies-Bouldin Index, and Silhouette Coefficient) are mathematically identical across both backends.

Visual Segmentation Comparison

Original Image



Classical Euclidean



Quantum-Inspired

- **Visual Parity:** Both solvers produce visually identical cluster assignments, confirming the functional correctness of the phase-space mapping.

Conclusions

- **High-Throughput Pipeline:** Developed a decoupled, multi-threaded C++/CUDA 5D image segmentation pipeline achieving over 120 FPS.
- **Algorithmic Parity:** Proved exact functional, visual, and mathematical convergence parity between the quantum-inspired and classical solvers.
- **Zero Performance Overhead:** Demonstrated that quantum-inspired distance metrics (simulating state overlaps) can be simulated efficiently on local GPU hardware via SFU intrinsics (`__cosf`).

Publications & Academic Activity

- **Book Chapter (HAIS 2026):**
  Şoptelea, S., Kallay, P., Tudor, M., Lung, R.I. (2027). *A Variational Quantum Classifier Applied to Romanian Financial Data*. In: Hybrid Artificial Intelligent Systems. Lecture Notes in Computer Science, vol 16596, pp. 429–440. Springer, Cham. DOI: 10.1007/978-3-032-29292-6_35.
- **Ongoing Research:** A comprehensive Quantum Machine Learning (QML) literature survey in the works.

Future Work

- **Spatiotemporal Tracking:** Integrating temporal continuity constraints (e.g., optical flow) to stabilize cluster boundaries across streaming video frames.
- **Physical NISQ QPU Deployment:** Migrating from local GPU-simulated phase overlaps to physical QPUs (e.g., via CUDA-Q) to measure true quantum state overlap latency.

Thank You!

Questions & Answers